

ITCS 8169 Assignment 1: The CNN Challenge

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Code: https://github.com/JiaxiangCxx/ITCS8169_Assignment1

Final checkpoint: checkpoints/places365_resnet50_final.pt (in the repository, with the configs, split and logs of every run).

1. Final result

Test accuracy: 95.5% (382/400) with a single ResNet-50 pretrained on Places365 and fine-tuned on our data. The test set was used once, after the model was chosen.

Validation. The 2,400 training images were split once into 1,920 for training and 480 for validation (stratified, 30 per class, seed 0; splits/seed0_val20.json). Every run uses this split, and all choices were made on it. The selected checkpoint (epoch 15) gets **96.0%** on validation (461/480; 460/480 when re-evaluated in fp32, the precision used for the test result). One validation image is 0.2 points and the standard error is about 0.9 points, so differences of one or two images are treated as noise.

2. Recipe and what mattered

- **Model:** torchvision ResNet-50 [1] (23.5M parameters, 4.1 GFLOPs) with the CSAIVision Places365 weights [2] and a new 16-class linear head. The backbone is frozen for 2 epochs, then all layers are trained.
- **Data:** 224×224 RGB input. The 15 grayscale 15-Scene classes [3] are copied to three channels; Flower is the only color class. Training augmentation: random resized crop (scale 0.85–1) and horizontal flip. Evaluation: resize to 256, center crop 224.
- **Training:** AdamW [4]; learning rate 8×10^{-6} (backbone) and 2×10^{-4} (head); weight decay 0.01; cosine schedule over 35 epochs; batch 32; mixed precision; cross-entropy with label smoothing 0.05 [5]. Training stops after 10 epochs without improvement and keeps the best validation checkpoint (25 epochs, 2.8 min on one Quadro RTX 5000). I call this recipe R.

What mattered. (1) *Pretraining data.* With recipe R fixed, Places365 weights give 96.0% and ImageNet weights 92.1%. On the same 480 images, 22 were fixed and 4 broken (McNemar $p = 5 \times 10^{-4}$ [6]). All three Places365 seeds (96.0, 95.4, 95.8%) stay far above the ImageNet run. Places365 has classes close to ours (bedroom, kitchen, coast, highway, skyscraper), and even a Places365 ResNet-18 (94.2%) beats the ImageNet ResNet-50. (2) *Fine-tuning.* ResNet-18 gets 80.4% from scratch and 80.2% with a frozen ImageNet backbone, but 90.4% when fine-tuned (Fig. 1a). (3) *Depth.* With Places365 weights, ResNet-50 is 1.6 points better than ResNet-18 (three-seed mean) for $2.3\times$ the FLOPs. (4) *No gain:* strong augmentation (Section 4) and flip test-time augmentation (one image worse). The learning rates, frozen warm-up and label smoothing were not tested separately.

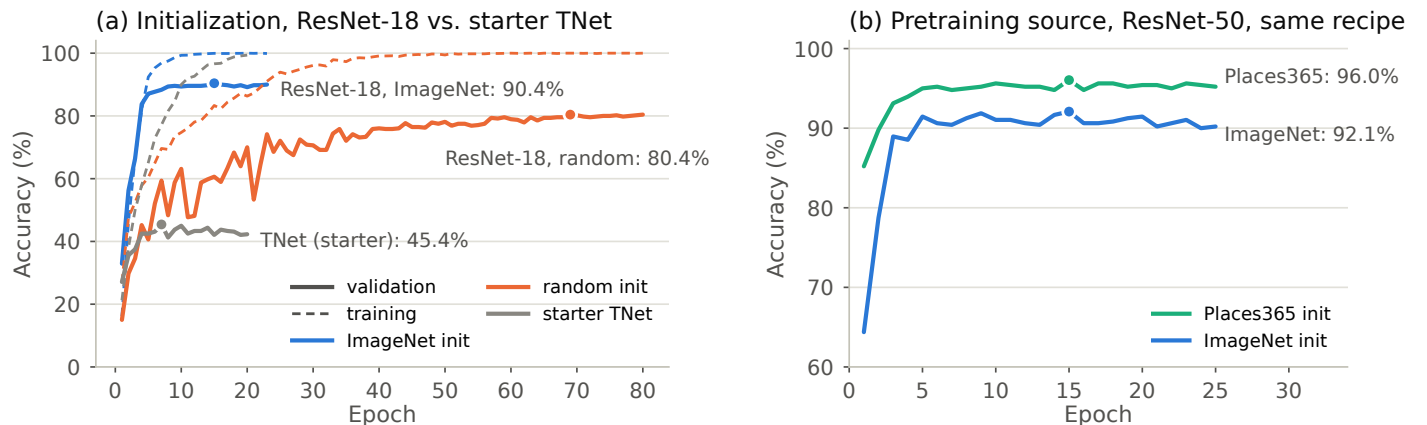


Figure 1: Accuracy per epoch on the validation split (dots: selected epoch). (a) Every model reaches about 100% training accuracy (dashed); they differ on validation (solid). (b) Same ResNet-50 recipe with different pretraining data.

3. Experimental journey

Experiment	Val. acc. (%)	Observation
Starter TNet (grayscale 64^2 , one conv layer)	45.4	99% train acc.; val. loss rises after epoch 6
ResNet-18, random init, RGB 224^2	80.4	Better, but a 20-point train–val. gap
ResNet-18, ImageNet, frozen backbone	80.2	Underfits (88% train acc.)
ResNet-18, ImageNet, fine-tuned	90.4	+10 points over both
+ strong augmentation	90.8	+2 images (noise)
ResNet-50, ImageNet, recipe R	92.1	Baseline for the next row
ResNet-50, Places365, recipe R (final)	96.0	Only the pretraining changed: +4.0 points
seeds 1 and 2	95.4 / 95.8	Three-seed mean 95.8
+ strong augmentation	94.8	–6 images
+ flip TTA (fp32)	95.6	–1 image vs. fp32 without TTA; not used
ResNet-18, Places365, recipe R	94.2	0.44× the FLOPs, 1.6 points lower

The first runs measured what pretraining and fine-tuning are worth on ResNet-18. Later runs changed one thing at a time: pretraining data, augmentation, test-time flipping, depth and seed. Seed 0 of the final recipe was selected before the seed runs and is the best of the three, so the three-seed mean (95.8%) is the fairer validation estimate.

4. Failure analysis: strong augmentation

Tried: a strong augmentation profile (crops down to 60% of the image, color jitter with $p = 0.8$, random grayscale, $\pm 10^\circ$ rotation) on ResNet-18 and on the final ResNet-50. *Why:* every fine-tuned model reached about 100% training accuracy while validation was much lower (90% for ResNet-18), which looked like overfitting. *Result:* ResNet-18 gained 2 validation images (noise). The Places365 ResNet-50 fell from 96.0% to 94.8% (–6 images, below all three seeds) and still reached 98–99% training accuracy. *Lesson:* 15 of the 16 classes are grayscale, so color jitter and random grayscale do nothing for them. Small crops can cut out the layout that identifies a scene (horizon, ceiling, room corners). The remaining errors are between similar scenes, which augmentation does not fix. Better pretraining closed the gap; extra regularization did not.

Test errors (18). Mountain→OpenCountry (3); Forest→OpenCountry, Kitchen→LivingRoom, LivingRoom→Bedroom and Store→Bedroom (2 each). Industrial is the hardest class (84%), with errors spread over four classes. Flower is 100% because it is the only color class, a property of the dataset rather than the model.

5. AI + human

I used OpenAI Codex and Claude Code (details in `AI_USAGE.md`). *Where AI helped:* Codex wrote the reproducible pipeline (fixed split, separate train/validation transforms, best-checkpoint logging), and Claude Code suggested comparing ImageNet and Places365 weights under one recipe. *Where I disagreed:* Claude Code recommended an ensemble of several models as the final model (about one point higher validation accuracy for about $27\times$ the compute). I chose the single Places365 ResNet-50.

6. Reproducibility

Seed 0, deterministic cuDNN, Python 3.13.9, PyTorch 2.8.0, torchvision 0.23.0, one Quadro RTX 5000. Re-running the first command reproduced the original run exactly (same per-epoch metrics and weights); the second reproduces the test result.

```
python train.py --config configs/best.yaml
python evaluate.py --checkpoint checkpoints/places365_resnet50_final.pt \
  --data-root dataset --output-dir runs/final_test_evaluation
```

References

- [1] K. He, X. Zhang, S. Ren, J. Sun. Deep residual learning for image recognition. CVPR, 2016.
- [2] B. Zhou, A. Lapedriza, A. Khosla, A. Oliva, A. Torralba. Places: A 10 million image database for scene recognition. IEEE TPAMI, 2018. Weights: http://places2.csail.mit.edu/models_places365/.
- [3] S. Lazebnik, C. Schmid, J. Ponce. Beyond bags of features: Spatial pyramid matching for recognizing natural scene categories. CVPR, 2006.
- [4] I. Loshchilov, F. Hutter. Decoupled weight decay regularization. ICLR, 2019.
- [5] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, Z. Wojna. Rethinking the Inception architecture for computer vision. CVPR, 2016.
- [6] Q. McNemar. Note on the sampling error of the difference between correlated proportions or percentages. Psychometrika, 1947.